TEMCO LEARN

T3000 API & Protocol

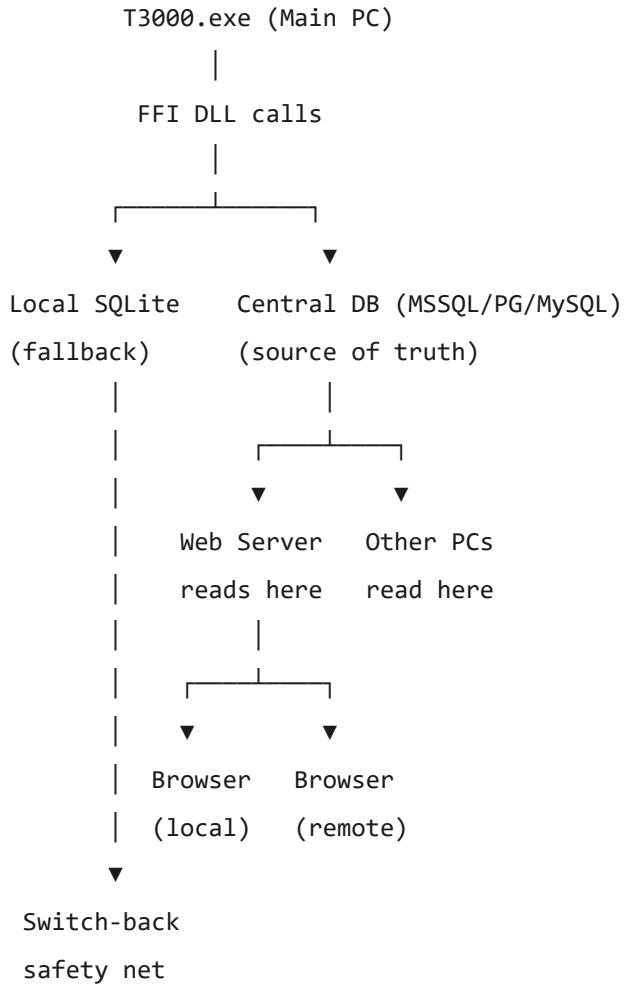
Multi-PC Data Flow Details · built Nov 13, 2026

Multi-PC Data Flow Details

> Per-table data flow reference for the centralized database multi-PC architecture. See centralized-database-multipc.md for the overall design.

1. Data Flow Diagram — Main PC

TEXT



FFI Write Flow (Main PC)

TEXT

```
Panel scan complete → FFI callback fires
  │
  ├─ Category A-C tables (DEVICES, INPUTS, OUTPUTS, VARIABLES,
  │   PROGRAMS, SCHEDULES, PID_TABLE, HOLIDAYS,
  │   TRENDLOGS, TRENDLOG_INPUTS, TRENDLOG_DATA):
  │
  │   ├─ 1. UPSERT to local SQLite      ← always first (fast, reliable)
  │   └─ 2. UPSERT to central DB        ← dual write
  │
  └─ Category D table (TRENDLOG_DATA_DETAIL):
      │
      └─ 1. INSERT to central DB ONLY   ← no local copy (saves GB)
```

Web Server Flow (Main PC)

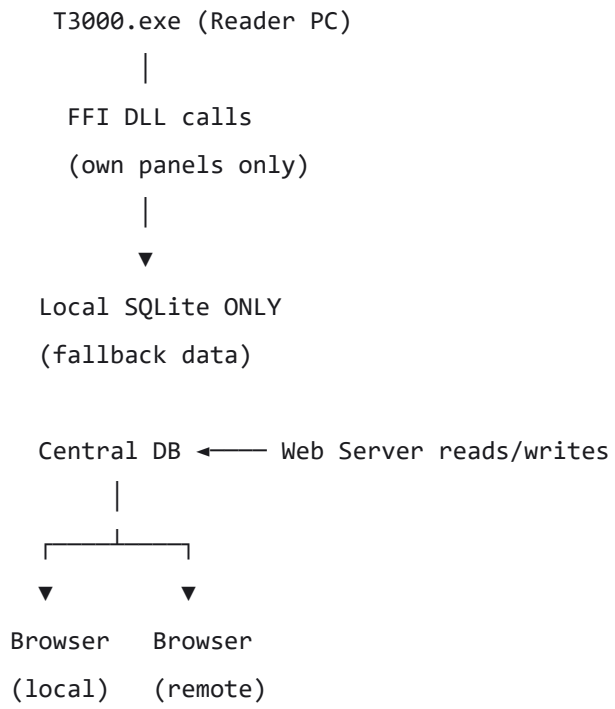
TEXT

```
Browser request arrives
  │
  ├─ READ request for data table (Cat A-F):
  │   └─ Query central DB → return results
  │
  ├─ WRITE request for data table (Cat A-F):
  │   ├─ 1. Write to central DB
  │   └─ 2. Write to local SQLite (dual-write for fallback)
  │
  ├─ READ/WRITE for local-only table (Cat G):
  │   └─ Use local_config_conn (always local SQLite)
  │
  └─ READ TRENDLOG_DATA_DETAIL:
      └─ Query central DB (only copy exists there)
```

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2. Data Flow Diagram — Reader PC

TEXT



FFI Write Flow (Reader PC)

TEXT

```
Panel scan complete → FFI callback fires
  │
  ├─ Category A-C tables:
  │    └─ UPSERT to local SQLite ONLY
  │         (does NOT write to central — main PC owns that)
  │
  └─ Category D (TRENDLOG_DATA_DETAIL):
       └─ INSERT to central DB ONLY
          (reader's own panels' trend data still goes to central)
```

Web Server Flow (Reader PC)

```
TEXT
```

```
Browser request arrives
  │
  ├─ READ request for data table (Cat A-F):
  │   └─ Query central DB → return results
  │       (sees ALL data from all PCs)
  │
  ├─ WRITE request for data table (Cat A-F):
  │   └─ Write to central DB ONLY
  │       (reader doesn't dual-write via web — only main does)
  │
  ├─ READ/WRITE for local-only table (Cat G):
  │   └─ Use local_config_conn (always local SQLite)
  │
  └─ READ TRENDLOG_DATA_DETAIL:
      └─ Query central DB
```

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3. Per-Table Detailed Flows

3.1 DEVICES (Category A)

Source: FFI `GET_PANELS_LIST` → DLL scans network for Temco controllers.

PC Role	FFI Action	Web Read	Web Write
Main	UPSERT local + UPSERT central	Central	Central + local
Reader	UPSERT local only	Central	Central only
Classic	UPSERT local	Local	Local

Conflict handling: UPSERT by `(serial_number)` or `(number, product_model)` — last write wins. Since each PC scans different panels, conflicts are rare.

3.2 INPUTS / OUTPUTS / VARIABLES (Category A)

Source: FFI `GET_PANEL_DATA` → reads point values from each discovered device.

PC Role	FFI Action	Web Read	Web Write
Main	UPSERT local + UPSERT central	Central	Central + local
Reader	UPSERT local only	Central	Central only
Classic	UPSERT local	Local	Local

Key: `(pid, panel_number, point_index)` — unique per panel, no cross-PC conflict.

3.3 PROGRAMS / SCHEDULES / PID_TABLE / HOLIDAYS (Category B)

Source: FFI reads these from panels along with point data.

Same flow as Category A. These are configuration data that changes rarely.

3.4 TRENDLOGS / TRENDLOG_INPUTS / TRENDLOG_DATA (Category C)

Source: FFI `LOGGING_DATA` cycle — discovery of trendlog configs and summary data.

Same flow as Category A. These are metadata/summary tables, relatively small.

3.5 TRENDLOG_DATA_DETAIL (Category D) ⭐ Special

Source: FFI `LOGGING_DATA` cycle — actual timestamped trend values.

PC Role	FFI Action	Web Read	Web Write
Main	INSERT central ONLY	Central	Central
Reader	INSERT central ONLY	Central	Central
Classic	INSERT local	Local	Local

Why special:

- Grows to millions of rows (one row per sample per trendlog per timestamp).
- Append-only (INSERT, never UPDATE).
- Keeping in local SQLite defeats the purpose of centralization.
- On switch-back, historical trendlogs are not critical for immediate operation.

3.6 UI/Config Tables (Category E) — GRAPHICS, ALARMS, etc.

Source: Web server only (user creates graphics, configures alarms via browser).

PC Role	FFI Action	Web Read	Web Write
Main	—	Central	Central + local
Reader	—	Central	Central only
Classic	—	Local	Local

3.7 Device Settings (Category F)

Source: Web server (user configures network/communication settings via browser).

Same flow as Category E.

3.8 Local-Only Tables (Category G)

Source: Internal application state.

Table	Always Uses
APPLICATION_CONFIG	Local SQLite
APPLICATION_CONFIG_HISTORY	Local SQLite
DB_BACKEND_CONFIG	Local SQLite
database_partition_config	Local SQLite
database_files	Local SQLite
database_partitions	Local SQLite
DATA_SYNC_METADATA	Local SQLite
TRENDLOG_DATA_SYNC_METADATA	Local SQLite

These tables are **never** written to or read from the central DB. They contain per-PC state that must remain local.

3.9 SYSTEM_LOGS (New Table)

Source: Application events, errors, audit trail.

PC Role	Write Destination	Read Source
Main (store_logs=1)	Central DB	Central DB
Reader (store_logs=1)	Central DB	Central DB
Any (store_logs=0)	Not written	—
Classic	Not applicable	—

4. Connection Resolution Logic

RUST

```rust
/// Determines which DB connection to use for a given operation.
fn resolve_connection(
    state: &T3AppState,
    table_category: TableCategory,
    operation: Operation,
) -> &DatabaseConnection {
    if !state.central_db_enabled {
        // Classic mode — always local
        return &state.local_config_conn;
    }

    match table_category {
        // Category G: Always local
        LocalOnly => &state.local_config_conn,

        // Category D: Always central when enabled
        TrendlogDetail => &state.t3_device_conn, // points to central

        // Categories A-F: Central for reads, dual for writes on main
        _ => match operation {
            Read => &state.t3_device_conn, // central
            Write => &state.t3_device_conn, // central (caller also writes local if main
        },
    }
}
```

5. Failure Scenarios

Central DB goes offline during operation

```
Write to central fails
  │
  ├─ FFI sync (Main):
  │    ├─ Local write already succeeded (done first)
  │    ├─ Log error, increment retry counter
  │    └─ Next sync cycle retries automatically
  │
  ├─ FFI sync (Reader):
  │    └─ Only writing local — unaffected
  │
  └─ Web server:
       ├─ Return 503 to browser with error message
       ├─ Log failure
       └─ UI shows "Central DB unavailable" banner
```

Switch from central to classic mode

```
Admin sets enabled=0 in setting.ini, restarts app
  │
  ├─ T3AppState.central_db_enabled = false
  ├─ t3_device_conn points to local SQLite
  ├─ All reads/writes go to local
  └─ Warning: TRENDLOG_DATA_DETAIL from central period is not in local
```

Network partition between PCs

```
Main PC and Reader PC can't reach each other
  │
  ├─ Both can still reach central DB → no impact
  ├─ Only Main can reach central → Reader falls back to local reads
  └─ Neither can reach central → both operate in degraded/classic mode
```